

Two Empirical Designs with Synthetic Court Data

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Purpose. This report presents a difference-in-differences estimation and a regression discontinuity estimation on two fictional datasets constructed for this exercise. Each dataset has a treatment effect purposely built into it; the goal is to recover that effect and to present the resulting estimates, tables, and figures clearly. All data, districts, and cases are fictional; results are a demonstration of estimation and reporting practice, not evidence about real courts.

Data

Court district data. We construct 40 fictional court districts. In 2019, 20 of them adopt a reform meant to speed up case processing; the other 20 never do. We observe all 40 districts every year from 2015 through 2022, tracking one outcome: the average number of days it takes to process a case in that district that year.

Fast-track case data. We separately construct 1,000 individual court cases, each assigned a continuous priority score between 30 and 70. Cases scoring 50 or above are eligible for a fast-track program. The outcome is the number of days it takes to process that case.

Summary statistics

Table 1: Descriptive statistics
Panel A. District-year panel, 2015–2022

	N	Mean	SD	Min	Max
Case-processing delay (days)	303	54.15	4.76	41.75	67.52
Reform district	320	0.50	0.50	0.00	1.00

Panel B. Individual cases, full sample

	N	Mean	SD	Min	Max
Case-processing duration (days)	940	57.35	8.36	30.37	81.60
Priority score	1000	50.02	11.90	30.03	69.88
Eligible for fast-track processing	1000	0.51	0.50	0.00	1.00

Notes: Panel A summarizes district-year observations for 40 districts from 2015 through 2022; 20 districts adopt the fictional reform in 2019. Panel B summarizes individual cases over the full priority-score range. Fast-track eligibility equals one for scores of at least 50. Statistics are unweighted and calculated using nonmissing observations for each variable.

Difference-in-differences

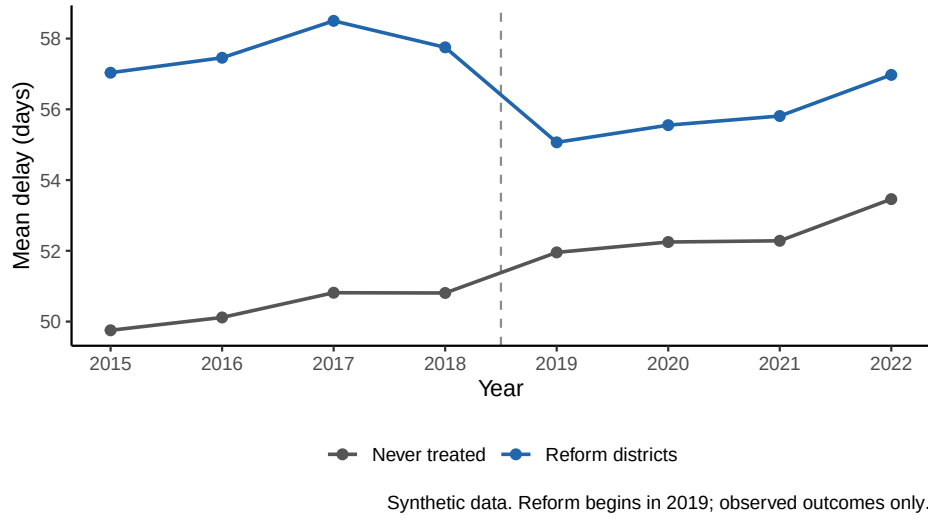


Figure 1: Observed mean delay by treatment group

Figure 1 shows broadly similar pre-reform movements, although reform districts experience a decline in 2018 that is not mirrored by the comparison group. This visual evidence is informative but cannot establish parallel untreated trends. In this simulation, that assumption holds in expectation because both groups share a common untreated trend and mean-zero shocks.

We estimate a model with district and year fixed effects, identifying the reform effect from the differential change after adoption. Standard errors are clustered by district to account for dependence across years. The model specification is given below:

$$Y_{jt} = \alpha_j + \lambda_t + \tau(Treated_j \times Post_t) + u_{jt}.$$

Table 2: District reform and processing delay

(1)	
Dependent variable:	Delay (days)
Reform district $\times$ Post-reform	-4.197*** (0.639)
District fixed effects	Yes
Year fixed effects	Yes
Observations	303
District clusters	40

Notes: OLS with district and year fixed effects, estimated by **regress**. Standard errors, clustered by district, are shown in parentheses below the coefficient. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Observations exclude missing outcomes.

The estimated reform effect is -4.197 days (95% CI $[-5.490, -2.904]$), close to and statistically indistinguishable from the built-in simulated effect of -4 days, as expected in this fully specified synthetic design.

Regression discontinuity

Fast-track eligibility is determined by whether a case’s priority score reaches 50. Because processing duration also varies with the score, comparing all eligible and ineligible cases would confound eligibility with systematic score differences. We instead compare cases near the cutoff, allowing the score–duration relationship to differ on either side.

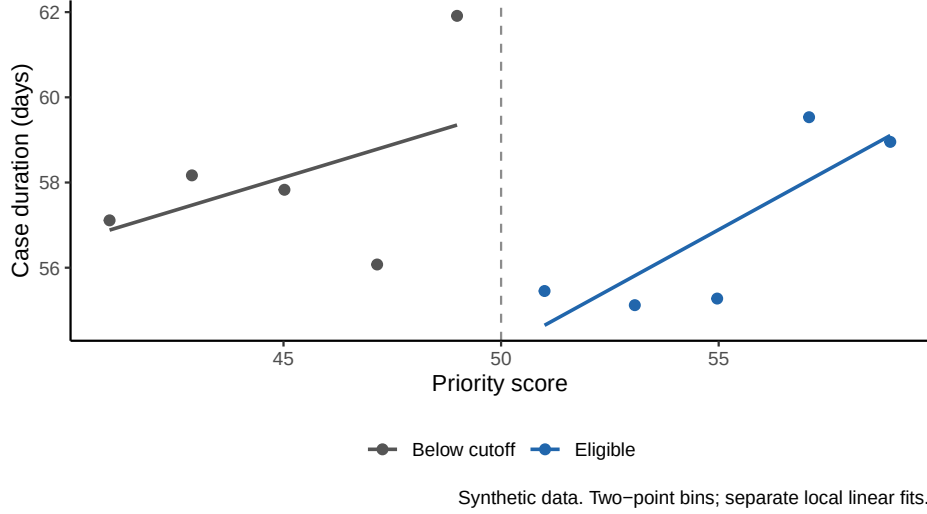


Figure 2: Binned outcomes around the eligibility cutoff

Figure 2 displays binned outcome means and separate fitted relationships around the threshold. The downward discontinuity motivates estimating a local treatment effect, but the outcome plot alone cannot establish the design’s validity. Identification requires potential outcomes to vary continuously at 50, without precise sorting or another intervention changing at the same threshold.

With $x_i = \text{Score}_i - 50$ and $D_i = 1\{\text{Score}_i \geq 50\}$:

$$Y_i = a + \tau D_i + bx_i + dD_i x_i + e_i, \quad |x_i| \leq 10.$$

Table 3: Fast-track eligibility and case duration

Dependent variable:	(1)
	Duration (days)
Fast-track eligibility	-5.574*** (1.481)
Bandwidth	10
Polynomial order	Local linear
Separate slopes across cutoff	Yes
Observations	457

Notes: Individual-level local linear OLS with separate slopes on either side of the cutoff, uniform weights, and a fixed bandwidth of 10 score points. Heteroskedasticity-robust standard error shown in parentheses below the coefficient. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. This is not bias-corrected RD inference.

Within the 10-point bandwidth, fast-track eligibility reduces estimated processing duration by 5.57 days (95% CI: -8.48 to -2.66); this is a local estimate at the cutoff.

Limitations. Identification is built into the simulation. Real policy research would require investigating assignment, missingness, measurement, spillovers, and specification sensitivity. The RD bandwidth is chosen for this demonstration, not selected from real evidence.